



NATIONAL SCHOOL OF BUSINESS MANAGEMENT

BSc (Hons) Computer Science – 17.2/18.1

4th Year 1st Semester Examination

22 May 2021

CS408.3- Embedded Systems

Instructions to Candidates

- 1) **Answer all questions.**
- 2) **Time allocated for the examination is three (03) hours and 30 minutes (Including downloading and uploading time)**
- 3) Weightage of Examination: 60% out of final grade
- 4) Download the paper, provide answers to the selected questions in a word document.
- 5) Please upload the document with answers (Answer Script) to the submission link before the submission link expires.
- 6) Answer script should be uploaded in PDF Format.
- 7) Under any circumstances E-mail submissions would not be taken into consideration for marking. Incomplete attempt would be counted as a MISSED ATTEMPT.
- 8) The Naming convention of the answer script – Module Code_Subject name_Index No
- 9) You must adhere to the online examination guidelines when submitting the answer script to N-Learn.
- 10) Your answers will be subjected to Turnitin similarity check, hence, direct copying and pasting from internet sources, friend's answers etc. will be penalized.

Question 1. (20 Marks)

Select the appropriate answer and under line.

1. Which protocol is used to link all the devices in the IoT?
 - a) TCP/IP
 - b) Network
 - c) UDP
 - d) HTTP

2. What is the bit size of the 8051 microcontroller?
 - a) 16-bit
 - b) 4-bit
 - c) 8-bit
 - d) 32-bit

3. The first page that you normally view at a website is its:
 - a) Home page
 - b) Master page
 - c) First page
 - d) None of the above

4. In reality, Internet protocol recognizes only
 - a) A location of the host
 - b) An IP address
 - c) A postal mail address
 - d) None of the above

5. Which is not one of the features of embedded devices?
 - a) Remotely controllable
 - b) Programmable
 - c) Can turn themselves off if necessary
 - d) All are features

6. Which one is not an element of embedded devices?
 - a) People
 - b) Process
 - c) Security
 - d) Things

7. The huge number of devices connected to the embedded devices has to communicate automatically, not via humans. What is this called?
- a) Internet of Services
 - b) Bot 2 Bot
 - c) M 2 M
 - d) Internet of content
8. A program written with the IDE for Arduino is called _____
- a) IDE source
 - b) Sketch
 - c) Cryptography
 - d) Source code
9. The monitoring of machines, gears and objects are achieved by which sensor?
- a) Humidity sensor
 - b) Proximity sensor
 - c) Touch sensor
 - d) Pressure sensor
10. Integrated development environment is
- a) Hardware application
 - b) Software Application
 - c) Communication application
 - d) Cloud platforms – Cloud IoT

Question 2. (20 Marks)

- a. What is an embedded system or Internet of Things (IoT) stands for? Explain with two examples.
- b. In Embedded Systems development process, Wireless Communications affect the development and implementation of the Embedded Systems? Explain how it could be affected.
- c. What are the important components of the Embedded Systems?
- d. What are the main challenges of the embedded Systems?
- e. Explain two industries can use embedded Systems platforms to improve their productivity?

Question 3. (20 Marks)

- a. What is a Satellite? Please explain.
- b. What is GPS? Please explain.
- c. What are the GPS signal errors, you will find when GPS signal travel from Satellite to device. Write four GPS Signal errors. Orbit error
- d. What are the main three segments used in GPS process? Explain.
- e. GPS is location based information providing device or a sensor. Explain three projects where you can incorporate GPS sensors for IoT development projects?

Question 4. (Marks 20)

Write short notes for following Quotations

- a. How do computers understand binary cords?
- b. What are active tags and passive tags in RFID?
- c. What are Internet and Protocols?
- d. What you will gain when you upgraded your system from 8bit and 16bit, explain?
- e. What is Microprocessor?

Question 5. (20 Marks)

You know that, Coronavirus disease 2019 (COVID-19) is an infectious disease caused by severe acute respiratory syndrome coronavirus. The viruses primarily spread between people during close contact, most often via small droplets produced by coughing, sneezing, and talking. In Sri Lanka in 2nd May 2020, detect the first COVID 19 patients and now it has increased up to 125,000 (+). As a software engineer, you have to designed application using embedded systems(IoT) for Health Care Sector? The Embedded Systems has many applications in the health care field, in both health monitoring, treatment and including tracking. You have to designed concept model, using Embedded Systems technology to automate the health care Sector. The project should submit with project proposal and it should consist with;

- a. Project description
- b. Problem statement
- c. Proposed solution
- d. Project execution and Methodology
- e. Conclusion

END OF THE PAPER